

## CREIL AP, France

WMO: 070570

Lat: 49.2522N

Lon: 2.5208E

Elev: 89

StdP: 100.26

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.9       | -4.0 | -8.8                          | 1.8 | -4.2 | -6.6 | 2.2 | -2.6 | 15.9                  | 10.5 | 12.4 | 9.8  | 1.5                   | 30   | 0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 30.2            | 20.0 | 28.1 | 19.3 | 26.1 | 18.4 | 21.0                | 27.9 | 20.0 | 26.2 | 19.1 | 24.6 | 3.5                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 13.6 | 23.2 | 17.8 | 12.9 | 22.2 | 17.0 | 12.3 | 21.3 | 61.0          | 28.1 | 57.5 | 26.2 | 54.5 | 24.7 | 25.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 11.2              | 9.1  | 7.8 | -9.0                       | 34.6 | 2.6                | 2.0 | -10.8                                              | 36.1 | -12.3      | 37.2 | -13.8      | 38.3 | -15.6      | 39.7 |    |    |
|                   |      |     | -9.3                       | 22.8 | 2.6                | 0.8 | -11.2                                              | 23.4 | -12.7      | 23.8 | -14.1      | 24.3 | -16.0      | 24.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.2   | 4.0   | 4.5   | 7.3   | 10.4  | 13.6  | 16.9  | 19.0  | 18.3  | 15.5  | 12.0  | 7.4   | 4.6  |
|                                                                                  | DBStd       | 6.34   | 4.20  | 3.94  | 3.50  | 3.43  | 3.26  | 3.14  | 3.04  | 2.74  | 2.83  | 3.54  | 3.60  | 4.05 |
|                                                                                  | HDD10.0     | 766    | 188   | 155   | 97    | 35    | 6     | 0     | 0     | 0     | 1     | 21    | 92    | 172  |
|                                                                                  | HDD18.3     | 2732   | 444   | 387   | 343   | 239   | 150   | 64    | 29    | 33    | 93    | 197   | 327   | 427  |
|                                                                                  | CDD10.0     | 1192   | 2     | 2     | 13    | 47    | 118   | 207   | 278   | 258   | 166   | 83    | 15    | 4    |
|                                                                                  | CDD18.3     | 114    | 0     | 0     | 0     | 0     | 4     | 21    | 48    | 33    | 8     | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 1197   | 0     | 0     | 0     | 13    | 57    | 231   | 470   | 341   | 80    | 6     | 0     | 0    |
|                                                                                  | CDH26.7     | 347    | 0     | 0     | 0     | 0     | 7     | 67    | 148   | 111   | 14    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.4    | 3.8   | 4.2   | 3.8   | 3.4   | 3.3   | 3.3   | 3.2   | 2.8   | 2.8   | 3.3   | 3.5   | 3.6  |
| Precipitation                                                                    | PrecAvg     | 663    | 54    | 46    | 43    | 46    | 58    | 59    | 62    | 47    | 61    | 56    | 68    |      |
|                                                                                  | PrecMax     | 979    | 136   | 99    | 161   | 120   | 147   | 120   | 145   | 129   | 119   | 142   | 105   | 159  |
|                                                                                  | PrecMin     | 480    | 4     | 8     | 3     | 3     | 3     | 16    | 16    | 9     | 1     | 20    | 10    | 20   |
|                                                                                  | PrecStd     | 114    | 28    | 27    | 33    | 28    | 30    | 27    | 34    | 31    | 29    | 28    | 24    | 34   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.5  | 15.8  | 20.1  | 25.1  | 27.5  | 32.0  | 33.5  | 33.1  | 28.8  | 23.9  | 17.4  | 14.3 |
|                                                                                  |             | MCWB   | 11.6  | 10.2  | 11.7  | 15.6  | 18.9  | 20.9  | 21.1  | 20.4  | 19.1  | 17.0  | 14.5  | 12.2 |
|                                                                                  | 2%          | DB     | 12.3  | 12.8  | 17.6  | 22.6  | 24.9  | 28.5  | 30.4  | 29.6  | 25.9  | 20.9  | 15.4  | 12.9 |
|                                                                                  |             | MCWB   | 10.7  | 9.4   | 11.3  | 14.3  | 17.1  | 19.6  | 20.0  | 19.6  | 18.3  | 16.4  | 13.3  | 11.2 |
|                                                                                  | 5%          | DB     | 11.0  | 11.4  | 15.2  | 19.4  | 22.7  | 26.0  | 28.1  | 27.0  | 23.3  | 18.6  | 14.0  | 11.8 |
|                                                                                  |             | MCWB   | 9.4   | 9.2   | 10.4  | 12.9  | 16.1  | 18.6  | 19.3  | 18.8  | 17.2  | 15.2  | 12.1  | 10.2 |
|                                                                                  | 10%         | DB     | 9.8   | 10.3  | 13.1  | 17.0  | 20.2  | 23.6  | 26.0  | 24.6  | 21.3  | 17.1  | 12.6  | 10.4 |
|                                                                                  |             | MCWB   | 8.4   | 8.4   | 9.6   | 11.7  | 14.8  | 17.4  | 18.6  | 17.6  | 16.1  | 14.4  | 11.1  | 9.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.2  | 11.2  | 13.2  | 17.1  | 20.1  | 22.6  | 22.1  | 22.0  | 20.5  | 18.1  | 15.4  | 12.7 |
|                                                                                  |             | MCDB   | 13.1  | 13.1  | 17.4  | 23.9  | 25.9  | 29.8  | 31.1  | 30.0  | 25.5  | 21.6  | 16.7  | 13.6 |
|                                                                                  | 2%          | WB     | 10.9  | 10.4  | 12.1  | 14.9  | 18.2  | 20.5  | 21.0  | 20.5  | 19.0  | 16.7  | 13.6  | 11.4 |
|                                                                                  |             | MCDB   | 12.0  | 12.1  | 15.5  | 20.9  | 23.3  | 26.5  | 28.2  | 27.6  | 24.1  | 19.8  | 15.2  | 12.6 |
|                                                                                  | 5%          | WB     | 9.6   | 9.5   | 11.1  | 13.3  | 16.6  | 19.2  | 20.0  | 19.4  | 17.9  | 15.7  | 12.4  | 10.4 |
|                                                                                  |             | MCDB   | 10.8  | 11.2  | 14.4  | 18.3  | 21.0  | 24.5  | 26.3  | 25.0  | 22.0  | 18.1  | 13.8  | 11.6 |
|                                                                                  | 10%         | WB     | 8.5   | 8.5   | 10.1  | 12.2  | 15.4  | 18.1  | 19.0  | 18.4  | 17.0  | 14.8  | 11.2  | 9.1  |
|                                                                                  |             | MCDB   | 9.6   | 10.1  | 13.0  | 16.3  | 19.3  | 22.6  | 24.9  | 23.2  | 20.4  | 16.7  | 12.6  | 10.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.1   | 6.6   | 8.7   | 10.3  | 10.2  | 10.5  | 10.7  | 11.0  | 10.6  | 8.1   | 6.0   | 5.3  |
|                                                                                  |             | MCDBR  | 5.2   | 8.2   | 12.8  | 14.4  | 14.0  | 14.7  | 14.7  | 15.7  | 13.9  | 9.6   | 6.5   | 5.3  |
|                                                                                  |             | MCWBR  | 4.4   | 5.8   | 7.4   | 7.5   | 7.2   | 7.0   | 5.9   | 6.9   | 6.9   | 5.7   | 4.9   | 4.5  |
|                                                                                  | 5% WB       | MCDBR  | 5.1   | 6.8   | 10.2  | 12.7  | 11.9  | 12.8  | 12.9  | 13.1  | 11.3  | 8.8   | 6.2   | 5.3  |
|                                                                                  |             | MCWBR  | 4.6   | 5.5   | 6.5   | 7.0   | 6.8   | 6.8   | 5.6   | 6.5   | 6.1   | 5.7   | 5.0   | 4.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.330  | 0.351 | 0.386 | 0.407 | 0.415 | 0.415 | 0.416 | 0.406 | 0.379 | 0.369 | 0.341 | 0.325 |      |
|                                                                                  | taud        | 2.394  | 2.355 | 2.247 | 2.210 | 2.225 | 2.257 | 2.254 | 2.305 | 2.367 | 2.385 | 2.403 | 2.369 |      |
|                                                                                  | Ebn at Noon | 715    | 787   | 814   | 837   | 845   | 846   | 839   | 830   | 817   | 755   | 693   | 662   |      |
|                                                                                  | Edh at Noon | 67     | 87    | 114   | 131   | 134   | 131   | 129   | 118   | 100   | 84    | 66    | 60    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.88   | 1.64  | 2.82  | 4.33  | 5.10  | 5.64  | 5.34  | 4.62  | 3.56  | 2.10  | 1.05  | 0.73  |      |
|                                                                                  | RadStd      | 0.10   | 0.25  | 0.35  | 0.50  | 0.51  | 0.53  | 0.47  | 0.46  | 0.30  | 0.17  | 0.10  | 0.11  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (40 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page